

THE ANH TRAN

ML System Engineer

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Education

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY (HUST)

HANOI, VIETNAM

Bachelor of Engineering in Computer Engineering — GPA: 3.88/4.0

Aug 2018 - Aug 2023

- **Ranking:** Top 01/176 of the graduating class.
- **Honors:** Talent Scholarship (7/8 semesters), Best Graduate Thesis (AIoT).

TRAN PHU HIGH SCHOOL FOR THE GIFTED

HAI PHONG, VIETNAM

Specialized in Mathematics, 9.0/10

Aug 2015 - Aug 2018

Working Experience

MOREH VIETNAM

HANOI, VIETNAM

AI System Engineer

July 2024 - Now

- **Moreh vLLM (Team of 5):** High-performance, AMD-optimized inference engine.
 - * **Business Impact:** Directly contributed to securing **\$15.6M in contracts** with leading enterprises in China and South Korea.
 - * **Achievements:**
 - Optimized inference performance to be **20-70% faster** than leading inference engines on the same AMD devices.
 - Achieved **90% of NVIDIA performance** while delivering significantly **better price per token (up to 2×)**.
 - Flagship models: DeepSeek V3/R1, GPT-OSS, and EXAONE 3.5; supporting for Llama 3.2/3.3 and Qwen 3.
 - * **Technical Stack:** LLM Serving Optimization, Distributed & Parallel Inference, LLM Benchmarking & Profiling.
- **Moreh LLM Eval (Solo Project):** **Moreh vLLM** Benchmarking & Evaluation Framework.
 - * **Business Impact:** **Reduced manual effort by 80%** (moving toward full automation) for benchmarking and evaluation per **Moreh vLLM** release, significantly accelerating the product development lifecycle.
 - * **Achievements:**
 - Standardized correctness evaluation and performance benchmarking and automated PDF report generating.
 - Developed S3-synced logging to enable simultaneous multi-instance runs and collaborative performance analysis.
 - * **Technical Stack:** LLM Evaluation & Benchmarking, CI/CD, Docker, AWS S3.
- **MoDiffusion (Team of 5):** Performance Optimization Suite for Diffusion Models.
 - * **Achievements:** Delivered significant inference speedups (5× on Motif, 2.2× on Stable Diffusion, 2.4× on Open-Sora), enabling AMD MI300X to match NVIDIA H100/TensorRT performance.
 - * **Technical Stack:** Model Compilation & Graph Optimization, Low-bit Quantization, Kernel Fusion, ONNX, TensorRT.
- **Moreh IR (Team of 5):** PyTorch-Compatible Compiler Layer of **MoAI Platform**.
 - * Developed and maintained **torch-2.1.0+moreh**, transforming PyTorch models into optimized computational graphs for high-performance execution on AMD backends.
- **Moreh Testbench (Team of 5):** Automated Quality Assurance Platform for **MoAI Platform**.
 - * Developed an automated daily benchmarking system, acting as the primary quality gate for **MoAI Platform** releases.

MB BANK

HANOI

AI Software Engineer

Mar 2024 - Jun 2024

- **MB Chatbot (Team of 2):** Enterprise RAG-based Customer Service System.)
 - * **Business Impact:** Serving **1000+ users** at MB247 department with **81% accuracy** and **sub-7-second response time**.
 - * **Achievements:**
 - Engineered a RAG system indexing 60,000+ multi-format documents using Milvus and Neo4j.
 - Developed an enterprise architecture integrating Python (AI Server) with Java Spring Boot (Gateway).
 - Implemented Redis caching and Keycloak authentication to ensure low latency and security.
 - * **Technical Stack:** RAG, LangChain, Milvus, Neo4j, Java Spring Boot, Redis, Keycloak.

ASILLA VIETNAM

HANOI

AI Engineer

Oct 2023 - Feb 2024

- **Ajikame - Hananoki Hospital Version (Team of 2):** Medical Behavior Detection.
 - * **Achievements:**
 - Respiratory Failure Detection: Developed a tracking system using Optical Flow and pose estimation to monitor breathing rates, achieving 70% accuracy on simulated data.
 - Epileptic Seizure Detection: Built a frequency-domain analysis module for pose data, achieving 65% recall and sub-30-second alert latency in crowded environments.
 - * **Technical Stack:** AI Camera Integration, Optical Flow, Pose Estimation, Frequency Domain Analysis.

VINBIGDATA

Teaching Assistance

HANOI

Jul 2023 - Sep 2023

- **Computer Vision:** Prepared exercise materials, supported lectures, and graded assignments for VinBigData employees.

PIXTA VIETNAM

AI Research Intern

HANOI

May 2022 - Nov 2022

- **Pixta Face Recognition (Solo Project):** Automated Facial Attendance System.
 - * **Business Impact:** Deployed a seamless **passive check-in** system for **50+ employees**, replacing manual methods with **100% precision** and **sub-1-second recognition time**.
 - * **Achievements:** Developed a high-performance system (real-time facial detection and sub-1-second facial recognition) supporting simultaneous passive check-in for up to 3 people, fully integrated with the 1Office platform.
 - * **Technical Stack:** Face Detection & Recognition, Real-time System Integration, PyTorch Lightning.
- **Pixta Image Embedding (Team of 3):** Foundation Model Research.
 - * **Achievements:** Developed a generalized backbone for downstream tasks by implementing Self-Supervised Learning (DINO, BEiT) and Vision-Language Pre-training (CLIP, FLAVA).
 - * **Technical Stack:** Self-Supervised Learning (SSL), Vision-Language Pre-training.

Honors and Recognitions

Honorable Mention, Hack (CX) Together 2025 (*HUST, Techcombank/2023*): Automatic Loan Approval (AI).

Best Presentation Award (*HUST/2023*): The Best Graduate Thesis (AIoT).

Talent scholarship, (*HUST/2020 - 2023*): Earned for 7/8 semesters.

Third Place, Chat'Innov Hackathon (*AUF/2023*): Laptop Advisor Chatbot (AI).

Third Place, Oraichain Hackathon (*Oraichain Labs, CYTAST/2023*): Decentralized Image Ownership Verification System (AI, Blockchain, Web3).

Honorable Mention, Vietnamese Traffic Safety Contest (*Vietnam Traffic Police, Dan Tri News/2022*): Traffic Density Monitoring System (AIoT).

Second Place, Hanoi Open Mathematics Competitions (*Hanoi Mathematical Society/2017*).

Technical Skills

Languages: Python, Java, Bash.

Frameworks & Libraries: PyTorch, vLLM, SGLang, LangChain.

Optimizations: torch.compile, ONNX, TensorRT, Triton.

Benchmarking & Profiling: InferenceMax, MLPerf, LM Eval, torch.profiler, nsight, rocprofv2, Perfetto.

MLOps & Workflow: Docker, AWS S3, Airflow, Git, Jira, CI/CD.

Databases & Vector Stores: Milvus, Redis, Neo4j, MySQL, MongoDB.

Backend & Tools: Spring Boot (Java), Flask & FastAPI (Python), Keycloak.

Extracurricular Activities

Teaching Assistant on Computer Vision, VinBigdata, 2023.

Head of Communication, BKFintech Club, HUST, 2023 - 2024.

Research Assistant, BKC Lab, 2023 - 2025.

Research Assistant, NCT Lab, 2022.

Language Proficiency

English: Professional working proficiency, TOEIC 805 (Nov 2022), IELTS 7.5 (Jul 2017).

Vietnamese: Native language.

References

Do Ba Lam

PhD, University of Vienna

Lecturer, School of Information and Communication Technology, Hanoi University of Science and Technology.

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